Test: Brier ALC by model and acquisition policy

algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.

Macro-task Brier ALC, lower is better. Bold: lowest among rows that never trained on these benchmarks.

model	random	uncertainty	conf. extremes	Bernoulli var.	exp. Brier red.
Constant (train mean)	**0.2472**	**0.2472**	**0.2472**	**0.2472**	**0.2472**
Empirical mean	0.2879	0.2881	0.2855	0.2887	0.2860
Beta-binomial	0.2494	0.2494	0.2478	0.2496	0.2474
ANP w64	0.2725	0.2710	0.2754	0.2775	0.2727
ANP w64 dense-prefix	0.2762	0.2732	0.2835	0.2787	0.2782
ANP w64 SAM	0.2663	0.2642	0.2718	0.2601	0.2651
ANP w64, random split*	0.2465	0.2445	0.2471	0.2438	0.2522
ANP w128, random split*	0.2515	0.2505	0.2505	0.2544	0.2575

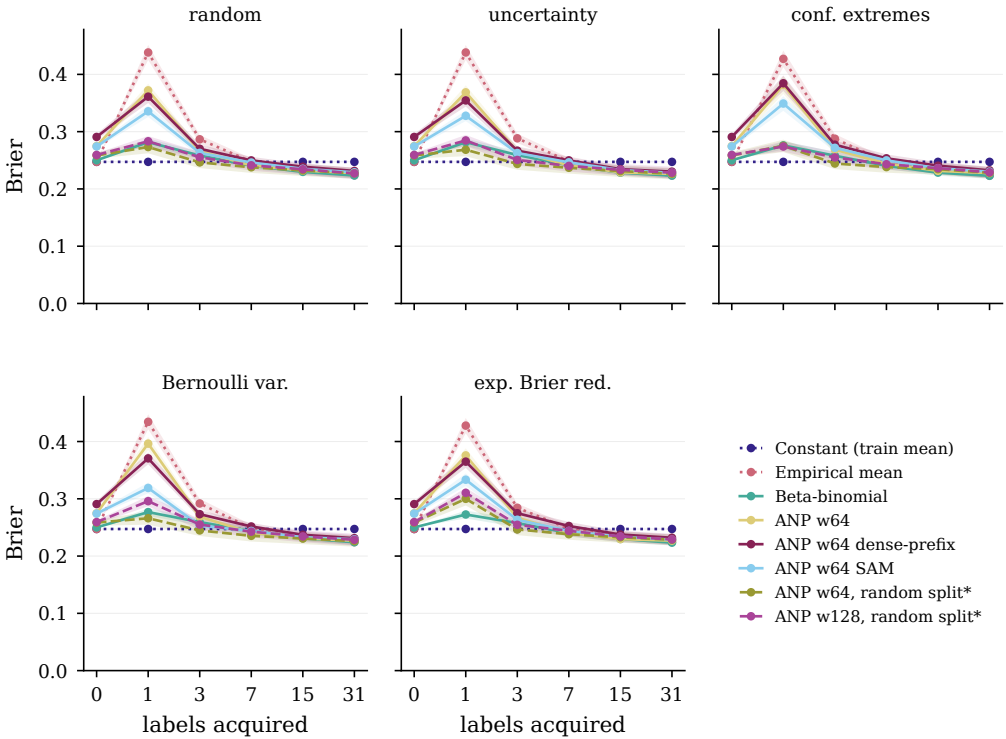
ALC minus beta-binomial under the same policy, paired over tasks: mean ± 95% CI. Bold: CI excludes 0.

model	random	uncertainty	conf. extremes	Bernoulli var.	exp. Brier red.
Constant (train mean)	-0.0022 ±0.0088	-0.0022 ±0.0091	-0.0006 ±0.0098	-0.0024 ±0.0091	-0.0001 ±0.0099
Empirical mean	**+0.0385 ±0.0030**	**+0.0387 ±0.0030**	**+0.0376 ±0.0033**	**+0.0390 ±0.0053**	**+0.0386 ±0.0040**
ANP w64	**+0.0230 ±0.0034**	**+0.0216 ±0.0033**	**+0.0276 ±0.0086**	**+0.0279 ±0.0059**	**+0.0253 ±0.0056**
ANP w64 dense-prefix	**+0.0267 ±0.0039**	**+0.0238 ±0.0039**	**+0.0356 ±0.0078**	**+0.0291 ±0.0062**	**+0.0308 ±0.0062**
ANP w64 SAM	**+0.0168 ±0.0042**	**+0.0148 ±0.0045**	**+0.0240 ±0.0075**	**+0.0105 ±0.0059**	**+0.0177 ±0.0066**
ANP w64, random split*	-0.0029 ±0.0187	-0.0049 ±0.0197	-0.0007 ±0.0196	-0.0058 ±0.0195	+0.0048 ±0.0192
ANP w128, random split*	+0.0020 ±0.0177	+0.0011 ±0.0191	+0.0027 ±0.0188	+0.0047 ±0.0191	+0.0101 ±0.0176

* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Test: Brier by labels acquired (mean ± 1 SE)

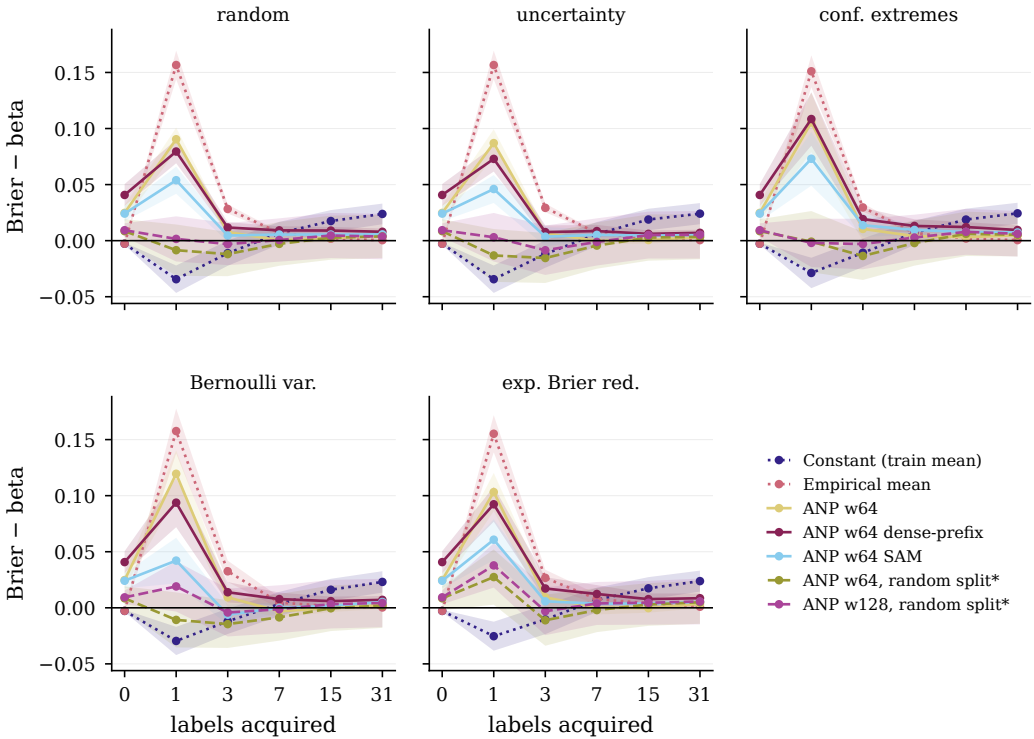
algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Test: Brier minus beta-binomial (paired, 95% CI)

algotune, cmmlu, editbench: no model trained on or was selected with these. 45 tasks x 10 order seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Validation: Brier ALC by model and acquisition policy

ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.

Macro-task Brier ALC, lower is better. Bold: lowest among rows that never trained on these benchmarks.

model	random	uncertainty	conf. extremes	Bernoulli var.	exp. Brier red.
Constant (train mean)	0.2525	0.2525	0.2525	0.2525	0.2525
Empirical mean	0.1892	0.1906	0.1944	0.1883	0.1921
Beta-binomial	0.1717	0.1726	0.1747	0.1704	0.1731
ANP w64	0.1735	0.1740	0.1759	0.1768	0.1749
ANP w64 dense-prefix	0.1735	0.1748	0.1718	0.1729	0.1719
ANP w64 SAM	**0.1667**	**0.1679**	**0.1677**	**0.1684**	**0.1708**
ANP w64, random split*	0.1519	0.1538	0.1536	0.1540	0.1611
ANP w128, random split*	0.1485	0.1503	0.1521	0.1528	0.1511

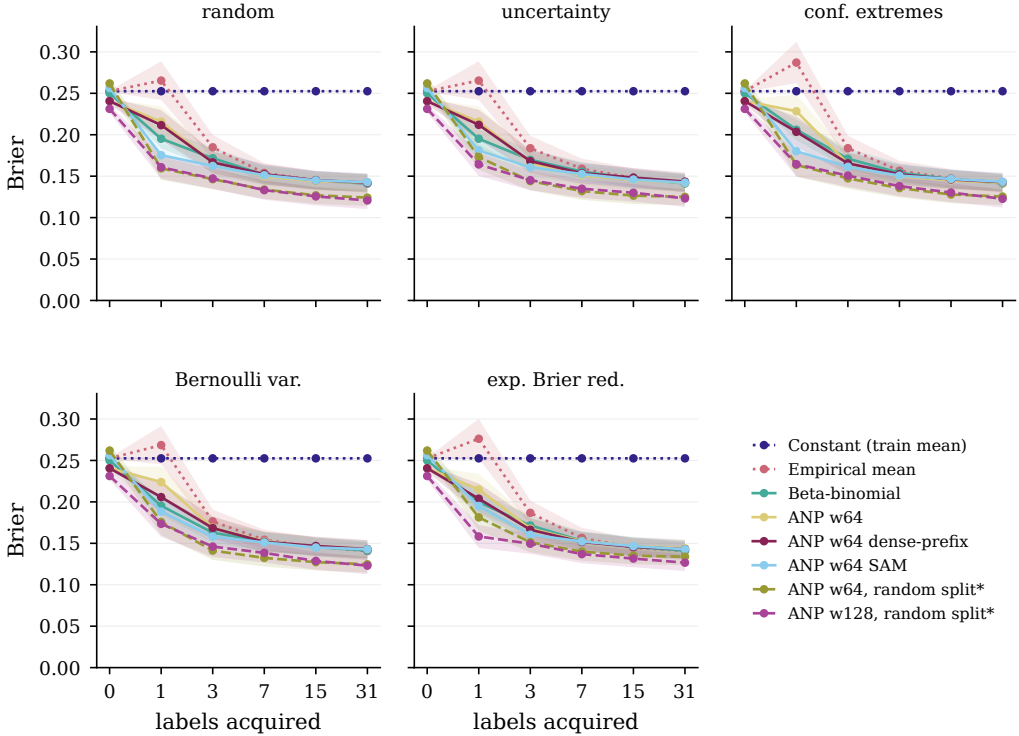
ALC minus beta-binomial under the same policy, paired over tasks: mean ± 95% CI. Bold: CI excludes 0.

model	random	uncertainty	conf. extremes	Bernoulli var.	exp. Brier red.
Constant (train mean)	**+0.0808 ±0.0211**	**+0.0799 ±0.0213**	**+0.0779 ±0.0214**	**+0.0822 ±0.0210**	**+0.0795 ±0.0212**
Empirical mean	**+0.0175 ±0.0056**	**+0.0180 ±0.0056**	**+0.0197 ±0.0058**	**+0.0180 ±0.0069**	**+0.0191 ±0.0088**
ANP w64	+0.0017 ±0.0034	+0.0014 ±0.0034	+0.0012 ±0.0043	**+0.0064 ±0.0060**	+0.0018 ±0.0064
ANP w64 dense-prefix	+0.0017 ±0.0044	+0.0022 ±0.0049	-0.0029 ±0.0067	+0.0026 ±0.0069	-0.0012 ±0.0066
ANP w64 SAM	**-0.0051 ±0.0039**	**-0.0047 ±0.0043**	**-0.0070 ±0.0050**	-0.0019 ±0.0057	-0.0023 ±0.0054
ANP w64, random split*	**-0.0198 ±0.0113**	**-0.0188 ±0.0115**	**-0.0211 ±0.0119**	**-0.0164 ±0.0119**	**-0.0120 ±0.0104**
ANP w128, random split*	**-0.0232 ±0.0110**	**-0.0223 ±0.0107**	**-0.0226 ±0.0115**	**-0.0176 ±0.0128**	**-0.0220 ±0.0093**

* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Validation: Brier by labels acquired (mean ± 1 SE)

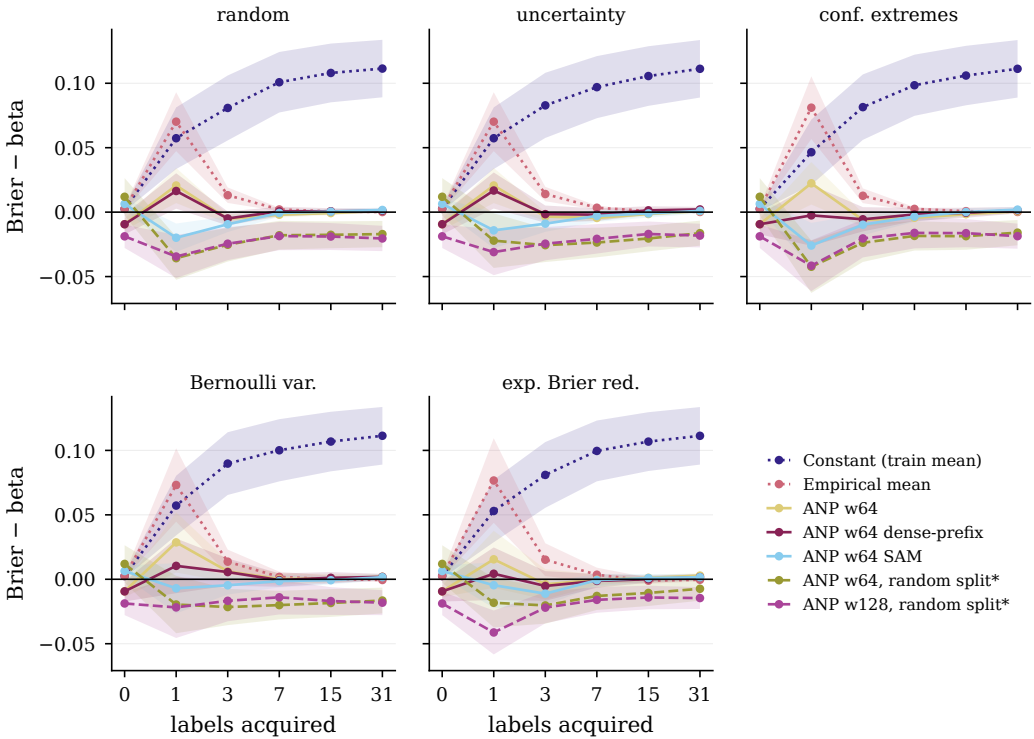
ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.



* random-subject split: trained on ai2d_test, bird_sql (validation) and editbench (test). Other ANPs used a benchmark holdout.

Validation: Brier minus beta-binomial (paired, 95% CI)

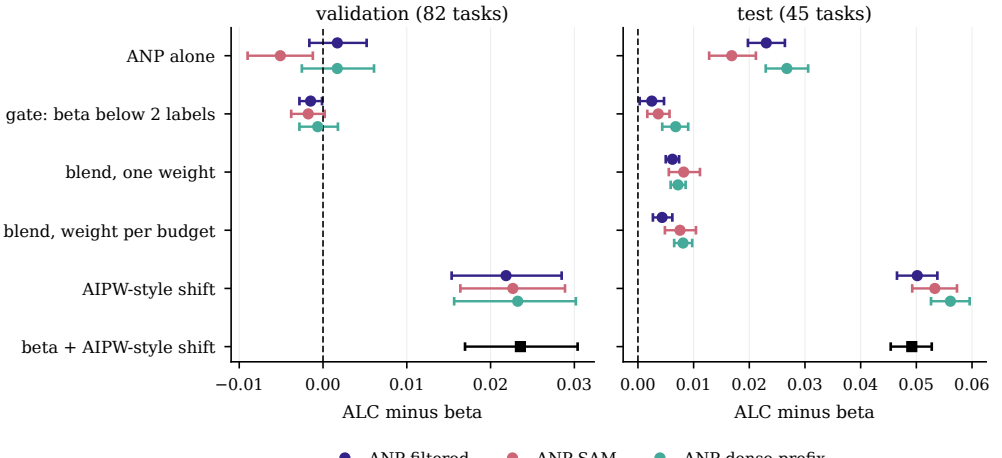
ai2d_test, bird_sql, financebench. ANP checkpoints were selected here, which flatters them. 82 tasks x 3 seeds.



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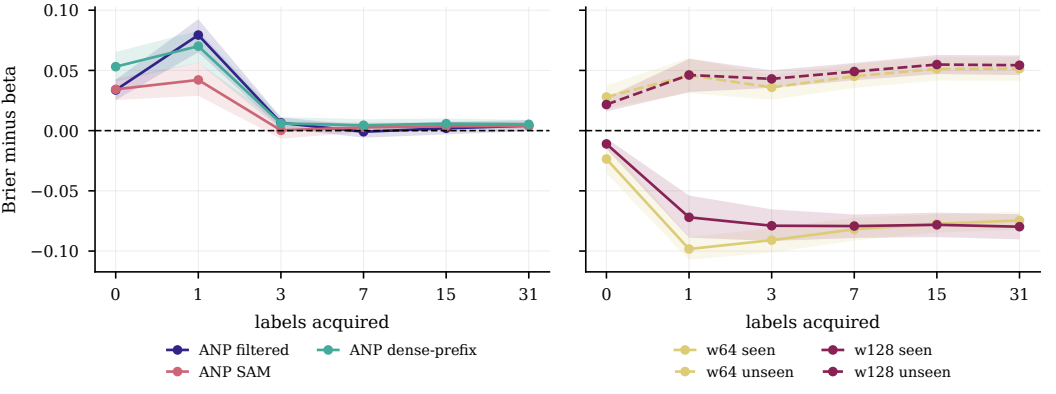
Can beta-binomial rescue the ANP?

Macro Brier-ALC minus beta-binomial, random acquisition; bars: 95% task bootstrap.
Negative = better than beta. Blend weights fitted on validation, so they appear on test only.



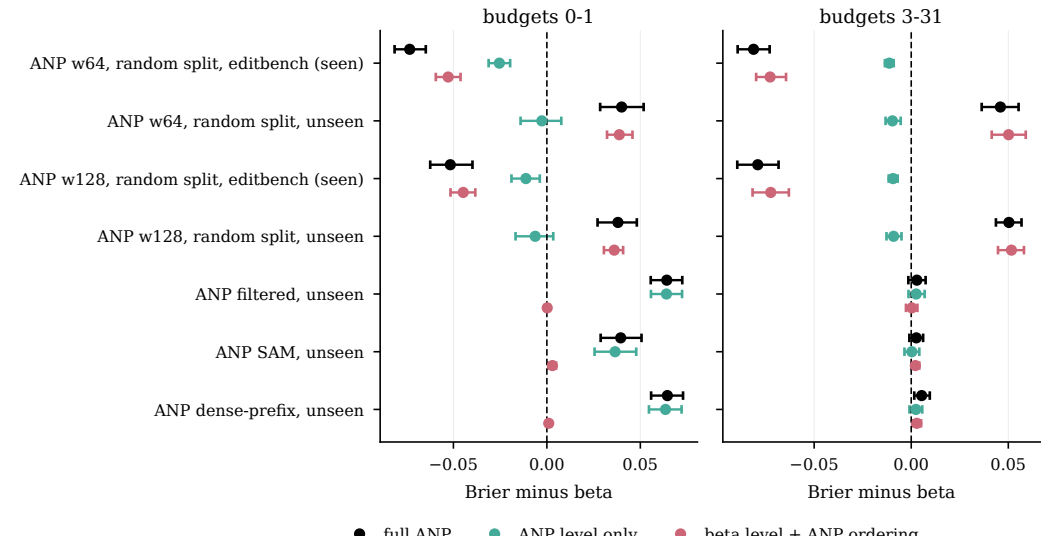
Where the ANP loses: Brier minus beta at each budget (test)

Left: benchmark-holdout ANPs on the three unseen test benchmarks. Right: random-split ANPs on editbench, which they trained on (solid), and on algotune + cmmlu, which they did not (dashed).



Which part fails on a new benchmark: level or item ordering? (test)

Level: the mean of the ANP's predictions for the task. Ordering: each item's offset from that mean.
Brier minus beta; negative = better than beta. Bars: 95% task bootstrap.



Does the ANP's loss grow with subject ability?

One point per task. Ability: the subject's accuracy on the task's acquisition pool (never scored).
Shaded: 5-95% of training-task accuracy (0.00-0.87). No test or validation subject appears in training.

